Smart Health 41 (2026) 100679

Contents lists available at ScienceDirect

Smart Health

journal homepage: www.elsevier.com/locate/smhl

A comprehensive review of wearable fall detection systems: Mechanisms, challenges, and future directions

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ARTICLE INFO

Keywords:
Fall detection system
Smartwatch
Accelerometers
Real-time monitoring
Gyroscopes

ABSTRACT

Specialized sensors are included in smart wearable devices engineered to provide effective human fall detection and continuous monitoring of routine activities. These devices utilize mainly gyroscopes and tri-axial accelerometers to record the patterns of body movement. The collected sensor data are processed by signal preprocessing pipelines to improve detection accuracy. The pipelines include noise reduction, feature extraction, and data normalization. In this study, a comprehensive survey of wearable fall detection systems is presented, focusing on the sensor, data processing methods, and fall detection techniques. This paper compares traditional threshold-based methods and state-of-the-art (SOTA) Machine Learning (ML) models, such as Convolutional Neural Networks (CNNs), Long Short-term Memory (LSTM) networks, and SOTA transformer models, in terms of detection performance, computational efficiency, and applicability to resource-limited wearable devices. It also highlights the importance of wireless communication technologies such as Bluetooth and Wi-Fi to send emergency notifications to caregivers or health care providers. But there are many problems, such as high energy consumption, delayed response, false alerts from normal daily activities, user variation, and an unbalanced data set. Moreover, this research focuses on emerging research trends and technical challenges in the design of real-time, reliable, and energy-efficient fall detection systems for the elderly and vulnerable populations. The outcomes of this study offer valuable insights into the efficiency of current approaches, shed light on limitations in current research, and propose future directions to improve wearable fall detection systems.

1. Introduction

Falls are a major public health issue, especially for older people. The World Health Organization (WHO) says that falls are the second most common cause of accidental deaths from injuries around the world. Older people are most likely to get hurt or die from falls (Tileria et al., 2020). Every year, about 684,000 people die from falls around the world. Millions more are hurt in ways that do not kill them but often put them in the hospital, make them disabled, and make their lives worse (World Health Organization, 2021). You can get hurt and lose money if you fall. Falls can break bones, especially in the hips, hurt the head, and make you scared of falling, which can make it harder to move around and be independent. It is hard for the economy to deal with it. Every year, falls among older adults cost the U.S. more than $50 billion in medical care (Florence et al., 2018). The problem is likely to get worse as the world's population ages. The number of people over 60 is expected to double by 2050. Unless good ways to stop and find them are put in place, this will lead to more falls and injuries and more problems for society (Ambrose et al., 2013). Finding falls early is very important to lower the number of injuries and deaths. The Personal Emergency Response System (PERS) and other old

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https://doi.org/10.1016/j.smhl.2026.100679
Received 14 February 2026; Received in revised form 15 May 2026; Accepted 13 June 2026
Available online 24 June 2026
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systems need the user to turn them on, so they do not work when someone is confused or unconscious. This shows how important it is to have fall detection that works in real time, automatically, and is not too obvious.

Smartwatches have built-in accelerometers and gyroscopes that identify fall events from sensor data due to sudden change in the posture of human. People look at these signals and try to figure out what kind of fall it was by using threshold-based methods or more advanced ML/Deep Learning (DL) models. You can work with data on your phone, computer, or in the cloud. Smartwatches can also use Bluetooth, Wi-Fi, and cellular connections to send automatic alerts to caregivers or emergency services. Smartwatches are a great way to keep an eye on falls in real time because they are small, easy to find, and can do a lot of things at once. This is especially true for apps that help older people and people who need health care from a distance (Bagala et al., 2012; Musci et al., 2020). Smartwatches are better than smartphones and single-use wearables at finding falls because they always keep the wrist in contact with the watch, sense accurately, and send alerts right away. Older adults are more likely to use and follow them because they know how to use them, can do a lot of things, and are easy to use. They have platforms that can change apps, add telehealth, and get updates powered by AI that are flexible and will work for a long time.

1.1. Motivation & problem statement

Falls are a leading cause of injury, hospitalization, and even death, particularly for older adults and individuals with mobility impairments. Many fall detection systems have been developed, but smartwatch-based systems have attracted much attention due to their portability, integrated sensing, convenience for users, and real-time monitoring ability (Bourke & Lyons, 2008; Tamura et al., 2009; Wang et al., 2020). However, there are still some important research gaps and practical challenges in current smartwatch based fall detection systems. The issues are summarized below:

- Limited sensor reliability and inconsistency in motion data acquisition under real-world conditions.
- Reduced detection accuracy caused by variations in user activities and complex daily movements due to lack of standardized datasets, evaluation protocols, and benchmarking strategies across studies.
- High false positive and false negative rates in threshold-based and traditional ML approaches.
- Increased computational complexity and energy consumption in DL-based systems, delayed response time and insufficient real-time performance in emergency situations.
- Privacy concerns and environmental dependency in camera-based monitoring systems.
- Insufficient discussion regarding practical deployment, user adaptability, and long-term usability of wearable devices.

Therefore, a comprehensive and structured review is required to critically analyze the existing fall detection technologies, sensing approaches, ML and DL methods, system architectures, and real-world implementation challenges. A such review can help to identify the limitations of current research, to highlight unresolved issues, and to provide future research directions for the development of more accurate, energy-efficient, reliable, and user-friendly fall detection systems.

1.2. Contributions

The main objective of this study is to present a detailed and systematic review of the smartwatch-based fall detection systems. The present review provides a critical review of existing methodologies, recent advances, algorithmic approaches and deployment challenges for smartwatch-enabled fall detection technologies. This review is focused on smartwatches with advanced sensing capabilities, which are portable, user-friendly, and suitable for real-time healthcare monitoring applications, unlike many existing studies that are generic in terms of wearable devices. The main contributions of this review are summarized below.

Technology Overview: Critical review of sensors integrated in smartwatches and their contribution in motion sensing, data acquisition and wireless communication for fall detection.

Algorithmic Approaches: A systematic classification and comparative evaluation of threshold-based, ML and DL methods used in smartwatch-based fall detection systems.

Datasets and Evaluation Metrics: Summary of popular datasets, wearable sensing modalities and major performance evaluation metrics such as accuracy, sensitivity, specificity and false alarm rate.

Challenges Identification: An examination of the primary technical and practical constraints such as sensor reliability, computation complexity, power consumption, privacy issues and real-world deployment challenges.

Future Research Directions: A review of emerging research opportunities such as context-aware systems, personalized models, sensor fusion, energy-efficient architectures and integration with smart healthcare ecosystems.

This review in general gives a systematic synthesis of the current literature, points out the existing gaps in knowledge and gives valuable insights for researchers, healthcare practitioners and developers in the field of wearable healthcare and intelligent fall detection systems.

In this paper, Section 2 describes the detail literature review, where as Section 3 elaborate different smartwatch technologies available for fall detection. In Section 4, different fall detection algorithms have discussed in detail. Section 5 describes the different available standard datasets used in fall detection and the popular evaluation metrices to assess the performance. Different challenges in fall detection based on smartwatch are presented in Section 6. Section 7 elaborates about current applications for fall detection and the limitations also of these available applications. Section 8 highlights the future research scope based on the identified challenges and the last section draw the conclusion of this study.

2. Literature review

Several relevant studies were thoroughly reviewed and organized the existing literature in a structured manner to facilitate a clear evaluation of different methodological approaches and their corresponding technical implications that clearly integrated the data sources and search strategy, inclusion and exclusion criteria, dataset selection, screening and selection process, data extraction and synthesis.

2.1. Methodological approaches

To ensure transparency and reproducibility, a systematic literature selection process was adopted for this review. The objective was to identify relevant studies focusing on fall detection systems, with particular emphasis on smartwatch-based and wrist-worn approaches.

2.1.1. Search strategy

The search included publications from 2010 to 2025 to include both early and more recent developments in the field. This study combines the keywords like these to do the search: "fall detection", "smartwatch", "wearable sensors", "activity recognition", "monitoring the elderly", and "devices worn on the wrist". Boolean operators (AND, OR) were employed to enhance the search and increase relevance (Snyder, 2019).

2.1.2. Inclusion and exclusion criteria

The process of selecting studies for the review employed established criteria to ensure that the included studies remained pertinent and of superior quality. This research included the studies that were published as peer-reviewed journal articles or conference papers, used wearable devices or sensors on smartwatches to detect falls, and showed experimental results or performance evaluations. This paper did not include studies that were not focused on fall detection, solely examined non-wearable methods such as standalone vision-based systems, or lacked sufficient details regarding their methodology or testing procedures.

2.1.3. Data selection

A comprehensive search was conducted across multiple scientific databases, including IEEE Xplore, Scopus, PubMed, and Web of Science. Among those, this study only focuses on the datasets which are used in wearable fall detection systems especially on smartwatch-based fall detection system.

2.1.4. Screening and selection process

The first search turned up a lot of articles. Titles and abstracts of the articles are examined to eliminate the studies that were not useful. Then, full-text reviews were done to see if the criteria made someone eligible.

Redundant records are removed that are present in more than one database. The final set of selected studies was subsequently organized according to sensing modality, algorithm type, and device characteristics. The main focus was on using smartwatches in implementations (Mubashir et al., 2013).

2.1.5. Data extraction and synthesis

To enable a thorough review, pertinent data was gathered from each chosen study, encompassing the device type utilized (e.g., smartwatch, smartphone, or other wearable sensors), the sensor modalities employed (including accelerometer and gyroscope), the algorithm categories applied (such as threshold-based methods, ML, or DL techniques), alongside evaluation metrics and dataset attributes. The gathered data were systematically scrutinized and integrated to discern emerging trends, prevailing challenges, and research deficiencies in smartwatch-centric fall detection systems (Sucerquia et al., 2018).

2.1.6. Limitations of the selection process

Even if there have been efforts to cover everything, there are still certain limits. These include the potential for publication bias, the exclusion of articles in languages other than English, and the rapid emergence of new research, particularly in 2024–2025, which may not be sufficiently represented.

To address these shortcomings, this review may be revised in the future using automated or continuous literature tracking methodologies.

Table 1
Fall detection studies.

Paper	Sensors	Methodology with description	Outcome	Strength	Gap
Yacchirema et al. (2018)	Wearable & IoT devices	IoT-based system with big data analytics for elderly fall detection	Real-time monitoring and alerts via IoT	Scalable, integrates IoT/cloud	Limited to connectivity; latency issues
Galvão et al. (2021)	Accelerometer, gyroscope, video	Multimodal DL combining wearable and video data	High detection accuracy with multimodal input	Robust under different conditions	High computation; privacy concerns
Chen et al. (2019)	Smartwatch (accelerometer)	Threshold and ML using wrist data	Reliable fall detection with wearable	Portable, user-friendly	Wrist data alone may reduce accuracy
Kwolek and Kepski (2014)	Depth camera & accelerometer	Depth maps and wireless accelerometer fusion on embedded platform	Improved detection in indoor settings	Fusion improves accuracy	High cost; poor outdoor performance
Jain and Semwal (2022)	Wearable inertial sensors	DL with novel feature extraction for pre-impact detection	Early detection before impact	Pre-impact focus reduces injuries	Dataset-specific; generalizability unclear
Qian et al. (2022)	Wearable accelerometer	IoT-based real-time fall detection with smartphone/cloud link	Efficient real-time alerts	IoT integration; low power	Relies on network availability
Waheed et al. (2021)	Wearable accelerometer	Noise-tolerant DL model (NT-FDS) for wearable data	Robust detection under noisy data	Handles real-world noise	Computationally intensive
Zurbuchen et al. (2021)	Wearable accelerometer	Multi-class ML with sampling rate study	Identified optimal sampling rates	Improved energy-efficiency	Dataset size limited
de Sousa et al. (2021)	3D accelerometer	Pre-impact detection using subject height & accelerometer	Improved anticipation of falls	Height personalization	Requires calibration
Yu et al. (2021)	Wearable accelerometers	Hierarchical attention CNN on sensor data	High detection accuracy	Captures temporal features	Complex; high computation
Kulurkar et al. (2023)	Wearables & IoT	AI-based predictive system for elderly	Continuous monitoring in smart home	IoT integration	Needs large training data
Alarifi and Alwadain (2021)	Wearable IoT sensors	Optimized CNN with heuristic methods	Enhanced detection rate	Optimized for IoT	Requires large datasets
Salah et al. (2022)	Accelerometer	Edge-AI fall detection system	Real-time inference on edge devices	Low latency	Limited adaptability
Xiaoqun Yu et al. (2021)	Inertial sensors	KFall dataset & benchmark algorithms	Public dataset for pre-impact falls	Benchmark for research	Limited diversity of subjects
Rezaei et al. (2023)	mmWave radar	Data-driven methods with radar signals	Contactless accurate detection	Privacy-preserving	Sensitive to environment
Wang et al. (2021)	Wi-Fi signals	Fine-grained indoor fall detection with Wi-Fi	Detects falls without wearables	Ubiquitous Wi-Fi use	Limited outdoor use
Shu and Shu (2021)	Multi-camera system	8-camera low-cost ML-based fall recognition	High accuracy in lab	Low-cost hardware	Scalability & privacy issues
Wang et al. (2023)	Shoes-floor nanogenerator	Motion recognition with energy-harvesting sensors	Self-powered detection	Energy-harvesting	Early stage; not wearable-friendly

(*continued on next page*)

Table 1 (*continued*).

Inturi et al. (2023)	Vision (skeleton keypoints)	LSTM using human skeleton features	Robust vision-based detection	Captures temporal patterns	Needs multiple cameras
Wang and Deng (2024)	Wearable sensors	Lightweight real-time fall detection algorithm	Efficient and reliable	Low computational load	Needs validation in real-world
Barshan and Turan (2023)	Wearable accelerometer	Double thresholding & fuzzy logic	Improved classification	Low-cost, interpretable	Limited adaptability
Mamchur et al. (2022)	Video stream	ML-based analysis of video	Accurate indoor detection	Simple deployment	Privacy & limited outdoor
Sadreazami et al. (2021)	RF/contactless	CNN with time-frequency analysis	Detects falls without wearables	Non-intrusive	Environment sensitive
Ramirez et al. (2021)	Vision (skeleton)	Skeleton-based ML for fall detection	Improved accuracy	Captures activity recognition	Requires good camera setup
Yao et al. (2022)	mmWave radar	Neural network & info fusion	High precision	Combines radar & fusion	High setup cost
Hassan et al. (2023)	Wearable sensors	Cost-effective fall detection with IoT	Affordable & scalable	Cost-efficient	May lack robustness in complex cases
Bach et al. (2026)	Accelerometer, gyroscope, inertial motion sensors	Lightweight edge neural network for fast fall classification.	Real-time fall detection with high efficiency and stable accuracy	Low latency, energy efficient, suitable for wearable/on-device use	Needs broader real-world validation; may be affected by wrist-motion variability
Manoj et al. (2025)	ECG, heart rate, temperature	Edge-AI for local ECG analysis, anomaly alerts.	Continuous health monitoring with rapid abnormal event notification	Low delay, privacy preserving, useful for remote elderly care	Primarily health monitoring; limited dedicated fall detection capability
Al-qaness et al. (2024)	Accelerometer, gyroscope, IMU sensors	Hybrid CNN-transformer for feature extraction and fall recognition.	Improved activity recognition and fall classification performance	Captures complex motion patterns, robust feature learning	Computationally intensive; requires optimization for smartwatch deployment
Zhengdong Li (2024)	Heart rate, temperature, motion sensors	User-centered wearable platform for continuous IoT health monitoring.	Effective real-time health tracking and alert communication	High usability, connected monitoring, supports long-term care	Focuses more on monitoring than fall detection accuracy; security concerns remain

2.2. Review of existing studies

A structured comparison of recent fall detection studies to illustrate the range of sensors, system architectures, and analytical techniques used in this domain is presented in Table 1. By including wearable, vision-based, and contactless sensing approaches, the table highlights how sensor selection affects detection performance, computational requirements, privacy, and deployment constraints. The summarized strengths and gaps reveal recurring technical challenges, such as false alarms, high computational cost, dependence on connectivity, environmental sensitivity, and limited scalability across users and settings. Overall, the table serves as a concise reference that supports comparative analysis, helps identify limitations in existing systems, and guides future research toward more reliable, efficient, and practical fall detection solutions.

Summarization of Fall Detection Studies, Key Strengths, and Research Gaps:

Now, the reviewed studies can be classified according to their sensing modalities like wearable sensor-based, vision-based, radar-based, Wi-Fi-based approach. As a result, this categorization provides a well structured understanding format about technological evaluation, methodological diversities, research discrimination for the fall detection system. In this scenario, Table 2 represents a comparative categorization of previous studies based on sensing technologies and deployment frameworks. This classification even makes it easier to identify current research trends, shortcomings, and practical deployment challenges associated with different fall detection system. Comparative study reveals that wearable systems based on smartwatches deliver a variety of advantages for real-time fall detection, such as continuous mobility support, direct body-contact sensing, low-latency monitoring, and better feasibility for common healthcare applications. As an outcome, wearable based techniques have turned out as a promising path for real-time and customized fall detection in smart healthcare settings.

Despite the advancements in the IoT-enabled, vision-based, radar-based, and hybrid fall detection systems, the current methods face major challenges such as false alarms, dataset diversity, privacy issues, computational overhead, connectivity dependence, and lack of real-world validation, as suggested by the findings of this study. Moreover, the absence of standard wearable based fall

Table 2
Summary of fall detection studies.

Category	Key references	Key strength	Research gap
Wearable sensor–based systems (accelerometers, gyroscopes, smartwatches, inertial sensors)	Chen et al. (2019), Jain & Semwal (2022), Waheed et al. (2021), Yu et al. (2021a), Barshan & Turan (2023)	Portable, affordable, user-friendly; effective real-time monitoring; suitable for elderly use	Reduced accuracy with single-sensor data, high computational cost in DL, limited dataset diversity
IoT and cloud-integrated systems	Yacchirema et al. (2018), Qian et al. (2022), Kulurkar et al. (2023), Alarifi & Alwadain (2021)	Real-time alerts, scalability, continuous monitoring across environments	Dependence on network connectivity, latency, and security/privacy issues
Vision-based methods (cameras, skeleton recognition, multi-camera setups)	Kwolek & Kepski (2014), Shu & Shu (2021), Ramirez et al. (2021), Inturi et al. (2023), Mamchur & Shakhovska (2022)	High accuracy in controlled indoor settings; rich contextual information	Privacy concerns, high cost, scalability issues, limited outdoor usability
Radar and RF-based systems (mmWave, Wi-Fi, RF signals)	Rezaei et al. (2023), Yao et al. (2022), Wang et al. (2021a), Sadreazami et al. (2021)	Contactless, privacy-preserving, effective in low-light	Sensitive to environmental interference, high setup/installation cost
Novel/Hybrid energy-harvesting and sensor approaches	Wang et al. (2023), de Sousa et al. (2021)	Innovative self-powered sensing, pre-impact detection	Early-stage, limited real-world validation, calibration needs
Benchmark datasets and evaluation studies	Yu et al. (2021b) (KFall dataset), Zurbuchen et al. (2021)	Provides standardized benchmarks, improves reproducibility	Limited subject diversity, constrained to specific environments

detection framework restricts the development of accurate, energy efficient and reliable real-time fall detection systems for practical healthcare deployment.

3. Smartwatch technologies for fall detection

Here, different most used sensor type technologies and different hardware and software platforms have discussed to do the processing of these sensor data. This section also describes various communication technologies which plays the key role for alerting the corresponding caregivers. At the end of this section various real time monitoring systems are explored.

3.1. Sensor types and capabilities

Smartwatches are equipped with a variety of sensors that enable them to monitor physical activity, detect sudden movements, and distinguish between different motion patterns. These sensors form the foundation of fall detection systems by capturing relevant biomechanical data in real time. The most commonly used sensors for fall detection include:

Accelerometer: The accelerometer is the most important sensor for fall detection because it registers abrupt movement changes due to falls, including impacts, free-fall, post-fall immobility and vertical drops (Bourke & Lyons, 2008; Kangas et al., 2008). It measures linear acceleration on the X, Y & Z axes. The total acceleration magnitude (A_{total}) is calculated using Eq. (1).

$$A_{\text{total}} = \sqrt{a_x^2 + a_y^2 + a_z^2} \tag{1}$$

where a_x, a_y, a_z represents the acceleration in three axes. A three-axis accelerometer can sense the static acceleration of gravity and the dynamic acceleration of motion, shock or vibration depicted in Fig. 1.

Gyroscope: A gyroscope measures angular velocity and tracks rotational movement and orientation. Combined with accelerometer data, it helps distinguish falls from activities like sitting, jumping, or bending, improving detection accuracy and reducing false positives (Musci et al., 2020).

Magnetometer: The magnetometer detects the Earth's magnetic field and is primarily used to determine direction and orientation. In fall detection, it is sometimes used to enhance sensor fusion techniques by correcting orientation drift from the gyroscope, though it is less central than accelerometers or gyroscopes.

Barometer: Some advanced smartwatches include a barometric pressure sensor to detect elevation changes, such as a sudden drop in height that might indicate a fall down stairs. While not always included, the barometer adds useful vertical context when analyzing fall scenarios (Patel et al., 2012).

Heart Rate Monitor (Optional Support): Although not a primary fall detection sensor, the heart rate monitor can provide supplementary information, such as detecting anomalies (e.g., sudden spikes or drops in heart rate) during or after a fall. This is particularly relevant in health-integrated systems aiming to assess fall severity or user condition.

Sensor Fusion and Data Processing: To make fall detection systems more accurate and reliable, they often use data from more than one sensor (sensor fusion). For example, combining data from an accelerometer and a gyroscope makes it easier to understand how falls happen, and adding contextual sensors like barometers makes it easier to detect falls in specific environments.

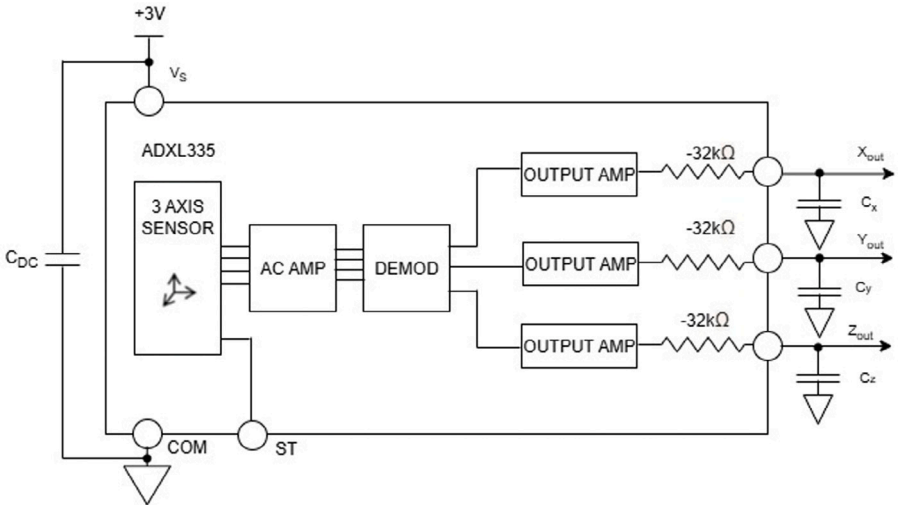


Fig. 1. Accelerometer.

3.2. Smartwatch-specific constraints and challenges

Some issues arise with smartwatch-based fall detection systems that do not occur with conventional body-worn or ambient sensing systems. When you wear a smartwatch on your wrist, it can easily pick up on the many and frequent movements of your arms that happen when you walk, eat, or gesture. This is not true for devices that are worn on the waist or torso. These movements often create motion patterns that look like falls, which increases the chances of false alarms (Mubashir et al., 2013; Shoaib et al., 2016).

Another important factor is orientation variability, which means that the smartwatch can move freely on the wrist. This can lead to different sensor readings for different people and situations (Casilari et al., 2017). Also, differences between the dominant and non-dominant wrist may affect how intense a motion is and how well it is detected, which is something that many studies do not pay enough attention to Bourke and Lyons (2008).

From a systems point of view, smartwatch platforms have strict hardware and software limits. For example, the battery capacity is small, the background sensing is limited, and the operating system has limits on continuous data acquisition (Sucerquia et al., 2018). Also, systems that use smartwatches need to find a balance between accuracy, energy efficiency, and user comfort. This is because constant sensing and processing can quickly drain the battery life (Sucerquia et al., 2018; Wang et al., 2019). So making smartwatches that can reliably detect falls requires a lot of careful study of both how people act and the limits of the devices themselves. This makes it a unique and difficult area of research (Nahian et al., 2026; Sharshar et al., 2023).

3.3. Hardware and software platforms

The sensors need to be good, but the hardware and software that process the data from the sensors, run the detection algorithms, and let people talk to each other also need to be good. If you want to make fall detection apps that can grow and work well, you need to know a lot about these platforms.

3.3.1. Hardware platforms

The hardware specs of smartwatches are very different, and this has a direct effect on how well they can detect falls in real time.

Processors (SoCs): Most modern smartwatches use low-power processors like those found in Qualcomm Snapdragon Wear, Apple S-series, and Samsung Exynos. These processors need to find a balance between how quickly they can do math and how much power they use, especially when the device is running ML algorithms.

Battery Life: The amount of energy used is a very important limit. When you watch and process data in real time, the battery can run out quickly. It is important to make algorithms that work well and use sensors that do not use a lot of power so that batteries last long enough for everyday use (Vavoulas et al., 2014).

Memory and Storage: Smartwatches have less RAM and space to store things than smartphones. This makes it hard to run DL models directly on the device because they are too big and hard to understand.

Sensors: As was said before, accelerometers, gyroscopes, and magnetometers are all common sensors. Some models also have extra sensors, like heart rate monitors and barometers. Different brands and models have different quality sensors and sample rates.

Connectivity: Most smartwatches have Bluetooth and Wi-Fi, but the best ones also have LTE/4G, which means they can work on their own. It is important to have a good connection so that caregivers or emergency services can get fall alerts and location information.

Table 3
Comparison of different software platforms.

Aspects / Features	Wear OS	watchOS	Tizen
Developer's Ecosystem and Platform	Developed by Google, based on Android	Developed by Apple, integrated with iOS ecosystem	Developed by Samsung, based on Linux-based operating system
Device Compatibility	Supports multiple smartwatch brands and third-party devices	Exclusive to Apple Watch devices	Mainly used in older Samsung Galaxy Watches
Development Environment	Compatible with Android Studio and Android development tools	Uses Xcode and Apple development framework	Uses Tizen SDK for application development
App Ecosystem	Wide range of applications and third-party integrations	Strong integration with Apple ecosystem and optimized apps	More limited app ecosystem compared to Wear OS and watchOS
Sensor Access	Provides access to smartwatch sensors and APIs	Supports advanced health and motion APIs such as HealthKit and CoreMotion	Supports sensor access through Tizen APIs
Background Processing and AI/ML Support	Supports background processing, real-time communication, Android AI, and cloud integrations	Efficient on-device processing with optimized power management and Core ML support for fall detection	Supports application background services with limited AI/ML support compared to newer Wear OS implementations
Health & Fitness Features	Health tracking through Google Fit and third-party APIs	Advanced health monitoring and fitness tracking capabilities	Basic health and fitness support
Strengths	Open ecosystem, flexibility, broad compatibility	Deep hardware–software integration and strong health features	Lightweight system with efficient performance on older devices
Limitations	Battery optimization varies across manufacturers	Limited to Apple ecosystem only	Reduced long-term support and shrinking ecosystem

3.3.2. Software platforms

The most popular wearable operating systems are Wear OS, watchOS and Tizen, especially in the smartwatch space. Wear OS is more open and compatible, watchOS has a better ecosystem and health integration. There are also sensor based applications on Tizen. Thanks to its optimized hardware-software integration, reliable sensor processing and advanced ML, watchOS is best suited among the three for health monitoring and fall detection apps (Privalova, 2019). Wear OS is a great platform to build custom wearable solutions, as it is highly flexible and scalable across multiple devices (Tileria et al., 2020). Performance is not consistent across manufacturers. Tizen has a lightweight performance and efficient access to sensors, but it has limits for the future development and long-term support for adoption (Patel & Mann, 2025). Table 3 shows the comparison of these software platforms.

Some manufacturers (e.g., Huawei, Fitbit) use custom operating systems. These may restrict third-party app development or limit access to raw sensor data, impacting the ability to implement custom fall detection solutions.

3.3.3. Development tools and APIs

Google Fit SDK (Astrup et al., 2016; Mishra, 2015), Apple HealthKit (Astrup et al., 2016; North & Chaudhry, 2016), and Samsung Health SDK (Astrup et al., 2016) provide access to sensor data and health-related metrics (Zhengdong Li, 2024; Yasmin et al., 2025). Platforms like Apple's Core ML, TensorFlowLite, and Edge AI SDKs allow for lightweight, on-device model deployment, enabling real-time fall detection without the need for cloud processing. Platforms like Firebase, AWS IoT, or custom backends support data syncing, model updates, and emergency alert services. To make fall detection systems that work well and are easy to use, you need to know what different hardware and software platforms can and cannot do. When choosing platforms for implementation, developers need to think about the trade-offs between performance, power use, and user experience.

3.4. Communication and data transmission

Efficient communication and data transmission are essential components of smartwatch-based fall detection systems. Once a fall is detected, timely alerting of caregivers, emergency services, or connected health systems is crucial to ensure a rapid response. The communication capabilities of a smartwatch directly influence the system's reliability, responsiveness, and integration into broader healthcare infrastructures.

3.4.1. Wireless communication technologies

Wearables communicate fall detection through various wireless technologies. Both Bluetooth and Bluetooth Low Energy (BLE) link up with smartphones to send data. BLE allows continuous monitoring in an energy efficient manner. Wi-Fi gives you direct access to the internet for cloud syncing and emergency alerts but it is more power hungry. Cellular (LTE/4G) can be used as a stand alone system that sends alerts and GPS information to caregivers or emergency services directly, but at a higher power and more expensive. NFC is a minor player mostly aiding in device pairing and user identification in health care emergencies.

3.4.2. Alerting mechanisms

As and when a fall is detected the system generates alerts to inform the user, caregivers or medical personnel. Local alerts are audio, visual, or vibration alerts on the smartwatch, often with a countdown timer to cancel false alarms. It is possible to configure it to send alerts remotely via SMS, calls, emails or push notifications, with information such as the GPS location and the time of the fall. In addition, cloud integration offers secure data storage, remote monitoring, long-term analysis, and integration with electronic health records for better healthcare support.

3.4.3. Security and privacy considerations

Fall detection systems works on health and location data with secure communication protocols. Encryption is generally done using SSL(TLS) and also applied to all transmission especially when the system sends any alerts or syncing data with cloud platforms. To prevent unauthorized access or false alerts, authentication mechanism is used that includes secure device pairing and user verification. Additionally, compliance with data protection regulations like HIPAA in the USA, GDPR in Europe, or equivalent national standards is essential to ensure the safe and lawful transmission of personal health information.

3.4.4. Challenges in communication

There are a lot of things that can make it hard for data to be sent and received quickly and easily during communication. When data transfer is delayed or fails, it can be hard to respond quickly to falls. Another concern is how much power the device uses, since constant communication or syncing data can quickly drain the battery. Also, network availability can be an issue because cellular or Wi-Fi connections might not work well in some places, like indoors or in rural areas. This can make it hard to send alerts on time. Smartwatch-based fall detection systems need to find a balance between how quickly they can send messages, how much energy they use, and how safe the data is to make sure that real-time alerts are useful and reliable.

3.5. Real-time monitoring and emergency response integration

As the fall detection feature is a real time system then smartwatch can able to communicate with emergency response systems without any difficulties. This thing ensures that the fall are not only detected but also cured instantly which can reduce the mortality rate.

3.5.1. Real-time monitoring

Smartwatch generally collect humans' activity data and look for patterns that are not usual by using various sensors. Complete the followings to recognize a fall:

Continuous Sensor Sampling: Getting data from the accelerometer and gyroscope in real time at the correct speeds (typically 50–100 Hz) so that fall-like occurrences are identified right away.

On-Device Processing: Many new smartwatches can do light processing right on the watch. They speed up response times and cut down on the requirement for outside devices by using built-in ML or threshold-based algorithms.

Immediate Feedback: When the watch thinks the wearer might fall, it normally sends them haptic or audible feedback. This lets them know and provides them 10 to 30 s to stop false alarms.

3.5.2. Emergency response integration

Detection alone is not good enough; a fall detection system has to react fast. Smartwatches use GPS to provide real-time location tracking to help responders find the user's location. "Sharing health data through platforms like Apple Health or Google Fit provides the medical context of age, history, heart rate, etc., and allows for faster, more personalized treatment in an emergency.

3.5.3. Communication protocols

Smartwatch-based fall detection systems use different ways to communicate to make sure that emergency responders can get to you quickly and effectively. One of the most common ways is SMS/call notification, where smartwatches automatically send emergency messages or calls to set contacts with information about the location and the incident. Professional caregivers and healthcare facilities use cloud-based dashboards to keep an eye on real-time data streams and incident logs. This lets them supervise from a distance and act quickly when needed. Third-party integration also connects some systems to national or private emergency medical response services through APIs or partnerships with mobile network operators. This lets you call an ambulance or a telehealth service directly when a fall is detected.

3.5.4. Challenges and considerations

There are a number of problems and things to think about when it comes to real-time monitoring systems for fall detection in smartwatches. Real-time monitoring systems by using smartwatches for fall detection have faced some obstacles. The most important concern is about reliability as these systems need to reduce communication delays, and any problems with the network can make it hard to send alerts on time. Battery life is also a problem because constant monitoring and connectivity, especially in LTE-enabled devices, can quickly drain power. The other concern is about false alarms, which requires a careful balance between sensitivity and specificity to stop people from responding to emergencies that are not really happening. Along with that, the privacy and user consent are very important because health and location data must be sent securely, with the user's clear consent and strict compliance with legal and ethical standards. Hence, the real-time monitoring and strong integration with emergency response turn smartwatches from passive trackers into active tools that can save lives. For them to be successful, they need quick, safe, and reliable ways to talk to each other.

Table 4
Fall feature threshold table.

Feature	Description	Typical threshold
Acceleration Magnitude	Detects impact or free fall	< 0.5 g or > 2.5 g
Angular Velocity	Captures rotation during fall	> 3 rad/s
Angle of Tilt	Measures body orientation change	> 45° post-fall
Inactivity Duration	Detects immobility after impact	> 2–5 s

4. Fall detection algorithms

Smartwatch-based systems work best when they have fall detection algorithms. These algorithms use data from motion sensors to figure out the difference between falls and normal daily activities (ADLs). There are four main types of these methods: threshold-based methods, ML approaches, DL techniques, and hybrid and ensemble methods. The following sections contain a short description of each method, its pros and cons, and any recent improvements made to those methods.

4.1. Threshold-based methods

Threshold-based methods are some of the oldest and most common ways to detect falls. They are easy to use, do not require a lot of processing power, and can be used in real time on devices with limited resources, like smartwatches. These methods rely on predefined thresholds applied to sensor data (primarily accelerometer and gyroscope readings) to differentiate between normal activities and falls (Barshan & Turan, 2023; Chen et al., 2019; Jiang et al., 2024).

4.1.1. Overview

Threshold-based methods keep an eye on certain features taken from motion data and send out an alert when these features go above or below certain important values. When someone falls, their total acceleration usually goes up quickly (for example, > 2 g or < 0.5 g during free fall).

A high acceleration value after a free fall could mean that the object hit the ground. A long-lasting change in orientation, as measured by a gyroscope or accelerometer tilt, can mean that the user has not healed after a fall. A period of immobility following a fall-like incident can further confirm an actual fall.

Table 4 shows the value of threshold and the corresponding falls depends on the feature.

Fall detection systems still usually employ threshold-based techniques as they are easy to use, require less computing power and can be implemented on wearable devices with limited resources to work in real time. However, the choice of the threshold is very context-dependent and cannot be generalized. The speed of movement of objects depends on the configuration of the sensors. For example, wrist worn smartwatches are more responsive to typical arm movements than sensors worn on the waist or torso. This means that systems that depend on smartwatches may not work well with thresholds set for other types of wearable technology. Also, the window size, signal filtering, and sampling frequency can change the peak values observed significantly. This may affect the fall detection accuracy. (Bourke & Lyons, 2008; Mubashir et al., 2013).

The threshold values also change depending on how many people are using the system and what is going on in the real world. A lot of studies use simulated falls by young people to set limits, but these might not work for older people, who fall more slowly and in more complicated ways. In smartwatch apps, fixed thresholds are even less reliable because of other issues, like the device's orientation changing and the fact that it can be worn loosely. So, you should think of threshold values as design parameters that can change, not as fixed values. Future research should focus on personalized or context-aware thresholding methodologies, along with hybrid systems that combine threshold-based techniques with ML models to enhance robustness and generalization (Casilari et al., 2017; Sucerquia et al., 2018; Vinston et al., 2026).

Threshold-based methods are still a big part of fall detection systems that use smartwatches, especially when there are not a lot of computing resources and things need to be done quickly.

4.1.2. Pros and cons

On low-power processors, this method works better for real-time execution. It is simpler to implement because it does not require labeled datasets or training. It is straightforward to verify or debug the behavior as it is predictable. This approach faces several limitations as well. The performance is mostly dependent on choosing the best threshold settings, which can differ from person to person. It may misclassify robust daily activities (for example, jogging or sitting fast) as falls while missing gradual or mild falls. Over time, thresholds do not adjust to user activity or context. For various age groups, body types, or activity levels, thresholds frequently require manual adjustment also.

4.1.3. Recent enhancements

To overcome their limitations, threshold-based methods are often enhanced through several methods. To cut down on false warnings, it blends several thresholds (such as acceleration + angle change + inactivity). Logical rules or decision trees used to refine detection. One of the best approaches is that building a hybrid model which integrates threshold-based detection with ML classifiers to achieve a balance between simplicity and accuracy.

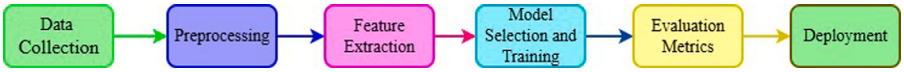


Fig. 2. Workflow diagram for ML approach.

4.2. ML approaches

ML has emerged as a powerful alternative to traditional threshold-based methods in fall detection systems, especially with the increasing computational capabilities of smartwatches. ML algorithms can learn complex patterns in motion data, enabling more accurate and adaptive detection of falls across diverse populations and activity types (Manoj et al., 2025).

4.2.1. Overview

Unlike rule-based systems, ML approaches leverage historical data to train models that can classify movements as either falls or non-falls. These methods rely on feature extraction, model training, and evaluation processes and are particularly effective in minimizing false positives and detecting subtle fall dynamics. The following workflow diagram Fig. 2 represents the whole process of ML-based fall detection methods.

Data collected during various tasks such as daily movements and fall simulation is obtained using standard smartwatch sensors such as an accelerometer and gyroscope. The noise is filtered by using existing low pass filters. The time series data is normalized and finally the feature is extracted from the collected raw data. Common features that can be extracted are mean, variance and standard deviation of acceleration, signal magnitude area (SMA), root mean square (RMS), peak values, zero-crossing rate and Fast Fourier Transform (FFT) coefficients. ML algorithms have been widely used to select and train models for fall detection. Support Vector Machines (SVM) are often employed for classes that are well separated. Generally Decision Trees and Random Forests are interpretable and robust. K-Nearest Neighbors (KNN) is simple to implement but can be computationally expensive. Naive Bayes is a light weight option and Logistic Regression is good for binary classification in linear separable data. The most common evaluation metrics used in fall detection are accuracy, precision, recall (sensitivity), F1 score and false positive rate (FPR). Depending on the hardware constraints and complexity, models can be deployed on-device (edge computing) or in the cloud.

4.2.2. Pros and cons

Learning of complex, non-linear patterns makes this method better than threshold method. Retraining it for various users enhances customization. It can differentiate between falls and similar activities (e.g., sitting down quickly) more efficiently. To increase robustness, it also integrates several sensor signals and features. This approach has also some drawbacks. It needs vast, labeled datasets, such as actual or precisely simulated falls. Particularly for deep models, training and inference can be resource-intensive, raising the computational cost. To avoid the risk of overfitting, appropriate training on particular topics or activities is necessary. The security of processing and preserving user motion data is one of the most delicate and concerning issues.

4.2.3. Recent enhancements

Recent trends in ML show a shift toward more efficient, adaptive, and real-time applications. Edge ML is becoming increasingly popular, with lightweight models such as TinyML or quantized neural networks enabling on-device processing without relying heavily on cloud infrastructure. Sometimes models are customized to fulfill users specific requirements as this performs better and real-world use cases are more accurate and useful. Many of us interested in hybrid methods (combination of ML and Threshold-based) rather than individual. These methods help you find a good balance between speed and accuracy, which is very important when you do not have a lot of money or time. Another big change is that more people are learning on the internet. In this kind of learning, models get new user data all the time. This lets them adapt to new situations and make changes. ML makes fall detection system more smarter and reliable when it embeds with smartwatch. It can find things that are personalized, flexible, and correct, but you need to plan it out carefully to get the right balance between accuracy, latency, and resource use.

4.3. DL techniques

DL has become very popular in research on fall detection because it can automatically learn and pull high-level features from raw sensor data. DL models do not need manual feature engineering like traditional ML models do. They can also model complex temporal and spatial dependencies in movement patterns that smartwatches record.

4.3.1. Overview

DL makes fall detection better by automatically learning patterns from unprocessed data. It does this by using recurrent and convolutional models to show how things change over time. It is more accurate than other methods, especially when working with big, varied datasets. So, this is a strong and reliable way to make fall detection systems that work in the real world. Table 5 shows different deep leaning models along with its operational area with brief description.

The available sensors like Accelerometer and gyroscope read data in X, Y and Z axis. Each sensor axis treated as a separate input channel (e.g., like RGB in images). Next these sensors data are segmented into overlapping and non–overlapping windows

Table 5
DL architecture for fall detection.

Architecture	Description	Use case in fall detection
CNN	Captures local temporal features from sensor signal windows	Effective for short-term motion analysis
RNN (Recurrent Neural Network)	Models sequential dependencies in time-series data	Captures the temporal evolution of falls
LSTM	A type of RNN that handles long-term dependencies and avoids vanishing gradients	Popular for long, noisy sensor sequences
GRU (Gated Recurrent Unit)	A lighter variant of LSTM with comparable performance	Suitable for resource-constrained devices
CNN-LSTM Hybrid	Combines spatial and temporal modeling	Highly effective for complex activity classification
Transformer Models	Use self-attention mechanisms to model dependencies over time	Emerging in real-time activity recognition applications



Fig. 3. DL training process.

(e.g., 2–5 s) which are fed into the model later. The workflow diagram Fig. 3 represents the whole training process of DL-based fall detection methods.

In the case of wearable based fall detection systems, the DL training includes the collection of accelerometer and gyroscope data and the labeling of the data as fall or non-fall activities. Thereafter, the data are pre-processed using normalization, segmentation, and noise reduction techniques to improve the signal quality. The dataset is then divided into training, validation and testing subsets to reduce overfitting and improve generalization. DL models (CNN, LSTM, or Transformer architectures) are trained with frameworks such as TensorFlow, Pytorch, or CoreML. Finally, model performance was evaluated using Accuracy, Precision, Recall, F1-score, confusion matrix, sensitivity, specificity, and false alarm rate for the assessment of real-world deployment.

Transformers are a DL architecture that outperform recurrent and convolutional sequence models by using an attention mechanism to analyze sequential data such as text, audio or time-series. This mechanism models the relationships between all elements in a sequence directly without recurrence and convolution (Paul & Phadikar, 2024; Yu et al., 2021).

In smartwatch-based fall detection, Transformer models leverage inertial sensor data from accelerometers and gyroscopes to recognize complex motion patterns. Raw time-series sensor signals undergo segmentation and feature embedding, followed by the addition of positional encoding to retain temporal order crucial for distinguishing falls from normal activities (Wang et al., 2020). Transformers work well because they have a self-attention mechanism that changes the importance of different time positions in a sequence. This lets the model focus on important moments in time, like sudden changes in speed or direction, while still keeping a general idea of the motion context. So, this means that you do not need to make your own thresholds as much. Compared to traditional ML and recurrent neural networks, Transformers are better at picking up long-range dependencies and complex motion transitions, which makes detection more reliable in real-world situations (Islam et al., 2024). Recent studies highlight the relevance of attention-based DL in wearable technology and healthcare sectors, particularly in activity recognition and anomaly detection (Abibullaev et al., 2023). Their ability to adapt makes them easy to incorporate into lightweight wearable systems, making Transformers a strong choice for fall detection in smartwatches (Amjoud & Amrouch, 2023; Kokab et al., 2022).

4.3.2. Pros and cons

DL works better than traditional ML methods, often getting better recall and accuracy. It shows that it is strong by being able to handle noise and changes in motion patterns better. It also helps build end-to-end systems, which make it easier to create models that can take raw input and make accurate predictions.

DL in fall detection faces several challenges, including high computational cost, as both training and inference demand substantial processing power and memory, making deployment on resource-limited devices like smartwatches difficult. These models also need a lot of labeled fall and non-fall data, which is hard to get in the real world. They also have a chance of overfitting, especially if they are trained on small datasets or in simulated environments. Also, because they use a lot of resources, they can drain the battery quickly, making it less useful to constantly monitor on wearable devices.

4.3.3. Recent enhancements

TinyDL and Model Compression: Quantization and pruning make it possible to run DL models on edge devices like smartwatches. This model lets the watch detect falls right away and send alerts right away, all while keeping your privacy without needing a strong internet connection. This level of efficiency keeps the watch's battery life intact and lets smart AI determine the difference between real falls and everyday arm movements. (Koo et al., 2023).

Table 6
Ensemble methods and applications in fall detection.

Technique	Description	Example use in fall detection
Bagging	Trains multiple classifiers on different subsets of the data and aggregates their results. Reduces variance.	Random Forest on extracted features
Boosting	Sequentially trains models, with each new model focusing on the errors of the previous ones.	Improves accuracy. Gradient Boosting Machines (GBM) on motion data
Voting	Combines predictions from diverse classifiers.	Majority or weighted voting using SVM, KNN, RF
Stacking	Trains a meta-model to learn how to best combine the predictions from base models.	SVM + LSTM + Logistic Regression ensemble

Transfer Learning: Transfer learning uses models that have been trained on huge datasets to accurately predict falls, even though a single smartwatch user usually only has a small amount of personal data. Also, it lets one model work on many different devices while still being very accurate, even when different smartwatch brands have different hardware sensors (Maray et al., 2023).

Personalized DL Models: Personalized DL models make fall detection much better by getting rid of "one-size-fits-all" algorithms. A personalized model learns the unique movement patterns of one user, while a generic model learns from data about the average person (Wu et al., 2015).

Federated Learning: Federated Learning makes it easier for smartwatches to detect falls by training models on each device without sending raw motion data. Only updates to the model are sent and combined to make a global detection model better while keeping user privacy (Haref et al., 2025).

Transformer-Based Approaches Transformer models enhance fall detection by employing self-attention to more effectively capture long-range temporal dependencies and intricate motion patterns (Vaswani et al., 2017). Using transformer models on data from wearable accelerometers and gyroscopes makes it easier to find falls, especially when the data is noisy, than using traditional DL methods (Guo & Nakayama, 2025b; Zhang et al., 2022). Transformer models are hard on computers, but lightweight hybrid architectures help smartwatches work by finding the right balance between speed and accuracy (Gomaa & Khamis, 2023; Guo & Nakayama, 2025a). Transformer methods look like they could work, but smartwatches need versions that are better for wearables and find the right balance between high detection accuracy, real-world efficiency, and low energy use.

DL can help make smartwatches' fall detection systems better. They could be more accurate, flexible, and easy to use in health monitoring systems that work in real time. But new ways are needed to make models work better, get data, and make them fit our needs.

4.4. Hybrid and ensemble methods

More and more researchers are using hybrid and ensemble methods to make fall detection systems that are more accurate, stable, and adaptable. These methods try to improve the system as a whole by using the best parts of different techniques, such as traditional rule-based systems, ML, or DL, to make up for their flaws (Al-qaness et al., 2024).

4.4.1. Overview

A. Hybrid Methods

Hybrid methods combine different algorithmic strategies into one system so that they can work together to get the best results. These systems are great for smartwatches because they help you make decisions quickly and use less battery power.

Here are a few common hybrid methods that are described in short.

Threshold and ML: A filter with a threshold looks at raw sensor data to find possible fall events. To figure out if the fall is real or not, ML classifier like SVM or Decision Tree are used to look more closely at these candidates. The good thing is that it only lets ML work on important events, which saves processing power and cuts down on false positives.

Rule-Based and DL: People usually use deep models like CNN and LSTM to figure out when someone is going to fall. Experts check predictions by setting rules, such as how long someone has been inactive or how they are sitting. It helps people understand things better and makes them more aware of the context.

Multi-Sensor Fusion: It helps us make better decisions by using data from a heart rate monitor, an accelerometer, and a gyroscope. Hybrid fusion methods combine data at the decision or feature level to give a clearer picture of what people are doing.

B. Ensemble Methods

When ensemble methods are used, then take predictions from more than one model and combine them to make an output that is more stable and accurate than what any one model could make on its own.

Types of Ensemble Techniques: Table 6 shows about detailed description of different Ensemble techniques along with the field where those are used.

Table 7
Comparative performance analysis of SOTA methods.

Method	Accuracy	Strength	Weakness	Best use case
Threshold	70%–85%	Fast	False alarms	Real-time low-power
ML	80%–92%	Balanced	Needs features	Smartwatch
DL	85%–97%	Learns features	Heavy compute	Cloud/offline
Hybrid	88%–98%	Best trade-off	Complex	Advanced systems

4.4.2. Pros and cons

Ensemble methods offer several advantages, including increased accuracy and robustness by combining the strengths of multiple models. They enhance generalization to new data, reducing the risk of overfitting, and provide the ability to integrate diverse perspectives from different models, leading to more reliable and effective predictions. Ensemble methods do have some issues, such as being more expensive to run because they use more than one model. They make implementation and tuning harder, which requires careful design and finding the best parameters. They might also add latency, which makes it harder to use them in real time on devices with limited resources, like smartwatches.

4.4.3. Recent enhancements

On-Device Hybrid Systems: Lightweight thresholding and a small ML model make sure that the system works quickly and efficiently on hardware with limited resources.

Cloud-Assisted Ensembles: Data that is raw or only partially processed is sent to a cloud platform where more powerful ensemble models look at it and send alerts back to the device or caregivers.

Personalized Ensembles: Ensemble methods that change over time based on the movement patterns of individual users, using online learning or federated learning frameworks.

Hybrid and ensemble methods are a promising area of research for fall detection. These methods combine the ease of threshold-based approaches with the intelligence of ML/DL models and the insights of many models to create a flexible, accurate, and efficient solution that is perfect for use in the real world on smartwatches.

4.5. Comparative performance analysis of SOTA methods

A review of recent SOTA methods shows that the performance of an algorithm depends a lot on how complex the data is, how limited the device is, and how the algorithm is used. Threshold-based methods are usually cheap and quick to respond, but they often give false positives because they are sensitive to how users move.

Support Vector Machines, Random Forests, and k-Nearest Neighbors are examples of ML methods that often show better robustness by learning how to find out the difference between different motion patterns from sensor data. DL models, especially those based on CNNs and LSTMs, often get better results on complicated datasets because they can automatically extract features and model time. Combining different strategies in a hybrid way usually gives the best performance because it combines efficiency with predictive accuracy Table 7 shows how the different methods compare to each other. The reported accuracy ranges are derived from existing studies in the literature (Bagala et al., 2012; Guo & Nakayama, 2025a; Musci et al., 2020; Paul & Phadikar, 2024; Wang et al., 2020). Actual performance may vary depending on factors such as dataset characteristics, sensor placement, evaluation protocols, and other conditions.

Interpretation of Comparative Results The design of fall detection methods determines which ones are better. Threshold-based methods work well for simple real-time applications because they do not need a lot of computation or training data. However, fixed rules make them less flexible (Bourke & Lyons, 2008). ML techniques enhance performance by discerning decision boundaries from motion features, facilitating superior differentiation between falls and routine activities (Zurbuchen et al., 2021). DL models often outperform conventional methods when large datasets are available, as they automatically learn complex spatial and temporal patterns (Galvão et al., 2021). Hybrid and ensemble approaches frequently achieve the best overall results by combining fast screening with refined classification Wang et al. (2020). Therefore, no single method is universally optimal.

5. Datasets and evaluation metrics

5.1. Datasets for fall detection

Publicly available datasets serve as a foundation for developing and comparing fall detection algorithms. These datasets typically consist of sensor data (e.g., accelerometer, gyroscope) recorded during both fall and non-fall activities.

Commonly Used Datasets: Several publicly available datasets have been developed to support research on fall detection using wearable and vision-based sensors. These datasets differ in terms of sensing modalities, participant demographics, data quality, and annotation detail. Table 8 provides a detailed summary of various datasets highlight the dataset's identity, purpose, sensing modalities, number of subjects involved, and any additional relevant information.

Real-life fall data is scarce due to ethical and safety constraints. Many datasets rely on simulated falls, which may differ in dynamics from genuine events (Nahian et al., 2026; Vinston et al., 2026).

Table 8
Recent available datasets in fall detection.

Dataset name	Description	Sensing modalities	Participants	Applicability	Data acquisition device
MobiFall (López Hincapie & Vargas Bonilla, 2017)	Contains data from falls and daily activities	Accelerometer, Gyroscope	11 subjects	Real smartphone/ wearable-based data	Smartphone (Galaxy S3)
SisFall (Sucerquia et al., 2018)	High-quality, labeled data for elderly fall detection	Accelerometer, Gyroscope	38 young & elderly	Includes many ADLs and fall types	Custom Embedded IMU (like smartwatch)
DLR Dataset (Songsheng Li, 2024)	Falls and ADLs recorded with a belt-worn sensor	3D Accelerometer	19 subjects	Balanced set of fall types	Specialized IMU
WEDA-FALL (Mubashir et al., 2013)	Annotated smartwatch fall dataset	Inertial sensors	Varies	Includes fall timestamps and segmentation	Fitbit Sense Smartwatch
UR Fall Detection	Video and sensor data (RGB-D & inertial)	Accelerometer, Kinect	6 subjects	Combines vision and wearable data	Kinect Cameras and x-IMU

5.2. Dataset characterization based on sensor placement and device type

In fall detection research, the characterization of datasets are strongly influenced by sensor placement and data acquisition devices, which directly affect signal quality, motion patterns and system performance. Therefore, a clear distinction between different dataset types is essential and relevance, particularly in a smartwatch-centric review.

Sensor placement strongly influences recorded signals in fall detection systems. Common locations include the wrist, waist/torso, and smartphone positions such as pocket or hand-held use. Wrist-based data from smartwatches often show high variability because frequent arm movements during daily activities may resemble fall patterns. In contrast, waist-mounted sensors better capture body center motion and often achieve higher accuracy under controlled settings (Casilari et al., 2017; Mubashir et al., 2013). Therefore, datasets from different sensor placements are not directly comparable, and performance results should always be interpreted according to sensor location.

Datasets also vary significantly based on the type of device used for data collection, such as Smartwatches, Smartphones, Dedicated wearable sensors (IMUs). Smartwatch-based datasets reflect real-world constraints, including limited sampling rates, energy consumption, and operating system restrictions on continuous sensing. In contrast, datasets collected using external IMU sensors often provide higher sampling precision but may not accurately represent real-world smartwatch deployment conditions (Demrozi et al., 2020).

This review emphasizes datasets collected using actual smartwatches, as they better represent realistic wrist motion, device limitations, and user interaction scenarios. However, many so-called wearable or wrist-based studies use prototype sensors instead of commercial smartwatches (Sucerquia et al., 2018). Therefore, datasets are classified as smartwatch-based, wrist-worn non-smartwatch sensor datasets, or non-wrist placement datasets.

5.3. Evaluation metrics

To assess the performance of fall detection algorithms, especially in imbalanced datasets (more non-fall activities than falls), a variety of metrics are used discussed in Table 9.

Where, TP: True Positives (correct fall detections), TN: True Negatives (correct non-fall detections), FP: False Positives (false alarms) and FN: False Negatives (missed falls).

Although accuracy, sensitivity, and specificity are commonly used in fall detection studies, they may not fully reflect the real-world performance of smartwatch-based systems. Wrist-worn devices create unique challenges that require broader evaluation methods. An important issue is the difference between window-level and event-level assessment. Many studies evaluate short signal windows, but a single fall can produce multiple overlapping windows, which may overestimate performance. Correctly classifying several windows from one fall does not always indicate reliable detection. Event-level evaluation, predicated on the identification of comprehensive fall incidents, offers a more accurate assessment of practical system efficacy (Mubashir et al., 2013; Noury et al., 2007). The false alarm rate is vital in smartwatch systems, as daily wrist movements may lead to false positives, undermining user trust and usability (Sucerquia et al., 2018). Simulated fall evaluations may not accurately represent real-world conditions, where erratic movements, ambient noise, and user variability frequently diminish system efficacy (Casilari et al., 2017). Recent studies underscore the significance of on-device evaluation, continuous monitoring, and user-centric metrics, thereby indicating the necessity for standardized protocols for smartwatch-based fall detection systems (Nahian et al., 2026).

Table 9
Evaluation metrics overview.

Metric	Description	Formula
Accuracy	Overall correctness of predictions	(TP + TN)/(TP + TN + FP + FN)
Precision	Proportion of predicted falls that are actual falls	TP/(TP + FP)
Recall (Sensitivity)	Proportion of actual falls that were correctly detected	TP/(TP + FN)
Specificity	Ability to correctly identify non-falls	TN/(TN + FP)
F1 Score	Harmonic mean of precision and recall	2 × (Precision × Recall)/(Precision + Recall)
ROC-AUC	Area under the Receiver Operating Characteristic curve	–
False Alarm Rate (FAR)	Number of false positives per unit time	FP/Time

5.4. Cross-validation approaches

Different algorithms and strategies are used in cross-validation methods for fall detection. Some of the methods used in k-fold cross-validation are SVM, Random Forest, and Neural Networks. They divide the data into k groups and use each group as the test set. Leave-One-Subject-Out (LOSO) is a common method used with CNNs, LSTMs, and Decision Trees. It checks each subject's data one at a time to see how well a model works on new data. Train/Test splits of 80/20 or 70/30 are quick ways to test, but they are not as good as LOSO or k-fold. People often use these splits with k-NN, Logistic Regression, and Gradient Boosting.

5.5. Challenges in evaluation

Testing fall detection is not simple. For example, the data is not evenly spread out, which can make it look like the accuracy is off. Also, simulated falls do not always match up with real falls, and there is not a standard way to compare different datasets and metrics. This makes it hard to trust comparisons between different methods. Require good data sets and ways to test them to make fall detection systems for smartwatches that work. In the future, the researchers should focus on getting real-world fall data that includes a wide range of people and making sure that all use the same methods to evaluate things. This will help make better solutions that more people can use.

6. Challenges in smartwatch-based fall detection

It is now easy and helpful to use a smartwatch to keep an eye on older people who might fall. It has some good points, but there are a lot of technical and people-related problems that make it less useful and harder to use in real life.

Sensor Limitations and Placement When you wear a smartwatch on your wrist, the motion data is not as accurate. Sensors worn on your wrist might not be able to record full-body kinematics when you fall, which could make them less accurate than sensors mounted on your torso. Also, sensor noise, drift, and misalignment make it even harder to process signals (Ghamari et al., 2016).

Power Consumption and On-Device Computation Smartwatches run out of battery quickly because they are always sensing, processing data, and talking to each other over the air. When you run ML or DL models in real time on limited hardware, you have to choose between performance and energy efficiency (Bach et al., 2026).

False Positives and Negatives It is hard to identify the difference between real falls and Activities of Daily Living (ADLs) because they both involve quick movements, like sitting down or jumping. This can cause either false positives, which send out alerts that are not needed, or false negatives, which do not catch real falls. Both of these things make people less likely to trust the system and make it less useful (Bourke & Lyons, 2008).

Lack of Real-World Fall Data Most models that recognize falls as these are trained on fake falls that healthy, young people usually do. This means that the models do not work for falls that happen by accident in real life, especially for older or weaker people (Sucerquia et al., 2018).

Personalization and User Diversity People move in different ways depending on their age, health, or how they walk. There probably will not be one algorithm that works for everyone, so there have requirement of some models that can be changed or made to fit each person (Patel et al., 2012).

Privacy, Comfort, and Compliance Some people might forget to wear the smartwatch, take it off because it hurts, or not want to use a device that always watches them, which is a privacy issue. These things make the system less reliable and less popular (Lin et al., 2023).

Connectivity and Emergency Integration Bluetooth or smartphone tethering is used by many smartwatch systems to send out emergency alerts. But if the connection is lost or cut off, it might take longer for caregivers or emergency services to get the alerts they need (Bagala et al., 2012).

Smartwatch-based fall detection is gaining popularity and shows great technical potential, but there are still major issues with sensor accuracy, real-time efficiency, personalization, and getting people to use it. People who make wearables, work with signals, do ML, and make healthcare policy need to work together to fix these issues.

7. Current applications and commercial solutions

Smartwatch-based fall detection systems is not only limited to research field but also now it is applied on real life. Now a days the system provides real-time monitoring and emergency alerts to help elder people, patients etc. These solutions use sensor technology, smart algorithms, and communication platforms to make people safer and more independent.

All of the commercial smartwatch-based fall detection features discussed in this section are backed up by official manufacturer documentation and publicly available technical sources to make sure they are correct and can be traced. Identifying specific model versions for each device, and the feature descriptions are based on the most recent information available at the time of writing.

7.1. Health monitoring platforms

Several mainstream smartwatches now offer integrated fall detection as part of broader health-monitoring features:

Apple Watch (Series 4 and above): The Apple Watch Series 4 and later models include built-in fall detection using accelerometer and gyroscope sensors, with automatic emergency alert functionality(verified April 2026). Upon detection, it prompts the user to respond, and if there is no response within about one minute, it automatically contacts emergency services and notifies designated emergency contacts. Additionally, it is integrated with Apple's HealthKit and Medical ID, enhancing its usefulness in health monitoring and emergency situations (Abou et al., 2022).

Samsung Galaxy Watch: The Samsung Galaxy Watch Active2/Watch 3 and later version provides fall detection features integrated with emergency SOS services(verified April 2026) (González-Cañete & Casilari, 2021). It also alerts pre-set emergency contacts and shares the user's real-time location, ensuring timely assistance during emergencies.

Withings ScanWatch: It offers fall detection along with ECG and blood oxygen monitoring, though fall detection may require pairing with smartphones as No official automatic fall detection feature was publicly documented as of April 2026.

7.2. Dedicated fall detection devices with smartwatch form factor

Some companies only deal with keeping an eye on the elderly and finding falls. They often make things easier for caregivers by providing simpler interfaces and backend support.

UnaliWear Kanega Watch: The Watch is made just for seniors and has features like voice help, automatic fall detection, and reminders to take your medicine. It does not need to be connected to a smartphone to work, and it uses AI to learn from how the user moves around every day.

SafeGuardian and Medical Guardian: SafeGuardian and Medical Guardian make pendants and watches that you can wear that can connect to cell phones and assess the fall. Most of the time, these devices come with subscription services that let you use monitoring centers that are open 24 h a day, 7 days a week. This way, you can always be safe and get help in an emergency.

7.3. Integration in healthcare and assisted living

More and more hospitals and other healthcare facilities are using smartwatches that determine the falls. They are very important for RPM programs for older people, help rehabilitation clinics keep an eye on post-operative or neurology patients who are at risk of falling, and are widely used in assisted living facilities, where alerts go directly to caregivers or nurses on site so they can act quickly.

7.4. Limitations of commercial solutions

Smartwatch-based fall detection systems are easy to find and useful for businesses, but they still have a lot of bugs that need to be fixed. People often get false alarms when they are just sitting around or working out. Alerts can also be late if the network is bad or the user does not respond. People with low incomes might not want to sign up either because they have to pay for the service every month. It is also hard to do fair comparisons and academic benchmarking because the algorithm details are not clear.

Adding basic AI to commercial smartwatch-based fall detection systems has made them safer and easier to use. They can also use the internet to keep people safe. But there are still problems with personalization, cost, and making sure the data is accurate. In these areas, businesses and academics need to work together and do more research.

Comparison of Current Smartwatch-Based Fall Detection Solutions

Comparison of available smartwatches based on their features are described in detail in Table 10. Generally features of the devices are enlisted as per the specifications reported by corresponding vendor. The features may vary as and when the firmware version, region, and subscription plan will be changed. All these features are verified on May 2026.

Table 10
Comparison of commercial smartwatch-based fall detection solutions.

Features \ Devices	Apple Watch (Series 4+) (Apple Inc., 2026)	Samsung Galaxy Watch (Samsung Electronics, 2026)	UnaliWear Kanega Watch (UnaliWear, 2026)	Withings ScanWatch (Withings, 2026)	Medical Guardian Watch (Medical Guardian, 2026)
Fall Detection Algorithm	Threshold & motion pattern matching	Threshold-based with sensitivity settings	AI-based adaptive learning	Basic threshold-based	Proprietary algorithm
Sensors Used	Accelerometer, Gyroscope	Accelerometer, Gyroscope	Accelerometer, Gyroscope	Accelerometer, ECG, SpO_2	Accelerometer, GPS
Smartphone Required	Optional (independent call/SOS on cellular models)	Required for full features	No (fully independent)	Yes	Optional (cellular option)
Automatic SOS Alert	Yes (calls emergency services if unresponsive)	Yes (sends location to emergency contacts)	Yes (calls monitoring center)	No (manual alert only)	Yes (calls 24/7 monitoring center)
Location Tracking (GPS)	Yes	Yes	Yes	No	Yes
Custom Emergency Contacts	Yes	Yes	Yes	Limited	Yes
Subscription Required	No (for fall detection only)	No	Yes	No	Yes
Battery Life	18–24 hours	24–48 hours	2–3 days	30 days	1–2 days
Target Users	General population, elderly	Active users, seniors	Elderly, fall-prone individuals	Health-focused users	Seniors, caregivers
Validation / Performance	Reported high sensitivity, limited transparency [1]	Performance varies, limited research	Claims high accuracy; no peer-reviewed data	Not validated in fall studies	Not peer-reviewed, vendor claims only
Limitations	Wrist-only data, limited adaptability	Some false alarms; dependent on phone	Costly subscription; less stylish	No automatic alert; limited data context	Subscription cost; generic algorithm
Model/Series	Series 4+ / SE / Ultra	Watch Active2 / Watch3 / Watch4 and later	Kanega Watch	ScanWatch 2	MGMove
Verified On	April 2026	April 2026	April 2026	April 2026	April 2026

7.5. Limitations in real-world deployments

People are buying more and more smartwatches that can identify falls, like the Apple Watch, the Samsung Galaxy Watch, and the Kanega Watch. But they have some big problems when you try to use them in real life.

High False Alarm Rate Many times, commercial devices mistake quick movements, like suddenly sitting down or moving your arms quickly, for falls. This could cause false alarms, which could make people turn off the feature. People who wear the Apple Watch say that fall alerts go off when they are working out hard or playing rough (Older People Who Are not Doing Well).

Underperformance in Elderly Populations Most commercial systems are set up with a small amount of data, and young adults usually pretend to fall. In real life, though, older people fall more slowly and in more ways, so the performance might not be as good. When trained in controlled settings, the algorithms do not always work well on real-life falls (Bagala et al., 2012).

Dependence on Network Connectivity Devices that use cellular or Wi-Fi networks, like the Apple Watch with LTE or Medical Guardian, may not send alerts on time if the user is in an area with poor signal coverage. This could mean that help takes longer to get there.

Battery Life Constraints Monitoring in real time and detecting falls use up battery life. For instance, the Apple Watch and Samsung Galaxy Watch usually only last one or two days and need to be charged a lot. This might not work for older people who have trouble with technology or who forget things.

Limited Transparency and Personalization Many commercial fall detection features are proprietary and use black-box algorithms that cannot be tested in public. Because it is not clear how it works, it is hard to detect how well it works or make changes to it to meet the needs of different users.

Cost and Accessibility A subscription is needed to use UnaliWear Kanega and Medical Guardian, for example. Seniors who do not have a lot of money may not be able to afford these services. A lot of people cannot use them because they cost too much and they have to pay for them every month.

Commercial solutions are easy to use, look nice, and have important safety features, but they do not always work well or adapt to the needs of different users in everyday, uncontrolled situations. You need to find a way to get around these real-world problems in order to make fall detection systems that work with smartwatches safe and easy to use.

8. Future directions and research opportunities

As fall detection using smartwatches continues to improve, future research needs to find ways to get around the problems with the current systems while also using new technologies. The following steps could help make fall detection systems more accurate, easier to use, and able to handle more people in real life.

Recent Advances in Smartwatch-Based Fall Detection Recent research shows a move toward real-world solutions that focus on smartwatches, with an emphasis on on-device intelligence and continuous monitoring for practical fall detection deployment. Recent studies have looked into lightweight DL and edge AI models for smartwatch hardware. These models can detect falls in real time while using less energy and computing power (Bach et al., 2026; Manoj et al., 2025). Transformer-based and hybrid models are also getting more attention because they can capture temporal motion patterns better than traditional methods (Al-qaness et al., 2024). Free-living datasets from commercial smartwatches make models more realistic and generalizable by capturing real behavior, noise, and long-term changes (Yasmin et al., 2025). Recent research focuses on user-centered metrics like false alarms, latency, and usability for real-world adoption (Zhengdong Li, 2024; Manoj et al., 2025).

Development of Real-World Fall Datasets Currently available dataset contain simulated falls which creates problems to perform accurately. So there have some definite requirement of real fall events data especially from older users. These data is very useful train and build better algorithm. It works with hospitals, nursing homes, and telehealth providers to set up data collection systems that are fair.

Personalized and Adaptive Algorithms From the past experience personalized thresholds and models learns and try to reduce the number of false positive and help to develop detection system more reliably. This changes over time by using ML on the device and continuous learning, such as federated learning.

Integration with Multimodal Sensing Using data from more than one source, such as heart rate, skin temperature, ECG, and audio from the environment, can help to identify the difference between real falls and movements that look like falls. It looks at low-power biosensors that are built into wearable devices and sensor fusion methods.

Edge AI and Energy Optimization Real-time decision-making and user privacy preservation can be achieved using lightweight approaches. AI models that run on the device (edge AI). To make batteries last longer without slowing down the device, future designs should focus on low-power architectures.

Real-Time Emergency Response Integration Using secure APIs and IoT frameworks to help emergency services, family members, or healthcare providers work together will speed up response times and make sure that medical help is always available when it is needed. It makes sure that everyone uses the same communication protocols and that smart home and healthcare platforms can work together.

Usability and Human-Centered Design If the interfaces are easy to use, the batteries last a long time, and the shapes are comfortable and not too noticeable, people will be more likely to follow the rules and trust you.

Clinical Validation and Regulatory Approval In order for fall detection to be a medical-grade tool, more research needs to be done on clinical trials, clear algorithm benchmarking, and following the rules set by the FDA and CE. It works with healthcare organizations to make sure that systems are up to clinical-grade standards.

Finding falls with a smartwatch is a new idea that brings together healthcare, AI, and wearable technology. To become a reliable, scalable solution for preventing falls and caring for the elderly, future research needs to focus on personalization, real-world data, and design that puts people first.

9. Conclusion

Wearable fall detection systems are an important step forward in healthcare technology, as they can quickly identify falls and trigger an emergency response using real-time sensing, continuous health monitoring, and artificial intelligence. This review extensively covered the underlying sensing technologies, smartwatch platforms, communication frameworks, and algorithmic approaches such as threshold-based, ML, DL, and hybrid methods employed in wearable fall detection systems. A comparative analysis of SOTA studies revealed major variations in detection performance, computational complexity, real-world reliability and energy efficiency, thus exposing key research gaps on false alarm reduction, dataset diversity, personalization and practical deployment on smartwatches.

The main point of this review is that DL-based multimodal methods are the best for finding falls. They are better than most other methods because they can model complicated relationships between time and sensors. This review says that threshold methods work well, but not as well in the real world, where things change all the time. ML finds a middle ground between accuracy and complexity. DL and hybrid models work best when things get tough. But the deployment of smartwatches still needs to be better for battery life, latency, and processing on the watch itself.

Researchers need to work on making models that are based on real-world fall data and thorough clinical testing, and that are personalized, energy-efficient, and easy to understand. New ideas in sensor fusion, edge AI, and user-centered design will help bridge the gap between academic prototypes and solutions that work well in the real world. In the end, wearable fall detection has a lot of promise. Not only can it make people safer, but it can also fit in perfectly with bigger healthcare systems, helping with proactive aging and ongoing care at home and in the hospital.

CRediT authorship contribution statement

Uttam Dolai: Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Bachchu Paul:** Writing – original draft, Supervision, Investigation, Conceptualization. **Utpal Nandi:** Writing – original draft, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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